CHITKARA UNIVERSITY

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNJAB

DEPARTMENT OF INTERDISCIPLINARY COURSES IN ENGINEERING

Digital Electronics and Computer Architecture Laboratory

Class Group: **15** Section: **B** Project Group: **1**

Project Abstract

Project Title: Temperature based fan speed control and monitoring with Arduino

Abstract: This project aims to develop a dynamic fan speed control and monitoring system driven by temperature variations, utilizing an Arduino platform. The system continuously monitors the surrounding temperature using a digital sensor (e.g., DHT11, LM35), and adjusts the fan speed accordingly to maintain desired thermal conditions. The fan speed is controlled via PWM (Pulse Width Modulation), allowing for precise adjustments based on the detected temperature thresholds. A key feature of this system is the real-time temperature and fan speed display on an LCD screen, providing continuous monitoring of the environment. By automating fan speed adjustments, the system enhances energy efficiency, reduces power consumption, and offers a reliable cooling solution for various applications, such as electronic devices, computer hardware, and climate control systems.

Application Area: **1**. Home Automation **2**. Office Buildings **3**. Industrial equipment **4**. Greenhouses

Impact on Society/Industry: **1.** Energy efficiency **2.** cost savings **3.** Improved Comfort **4.** Extended Equipment Lifespan **5.** Educational value

Sustainable Development Goals: 1.SDG 7: Affordable and Clean Energy **2.**SDG 9: Industry, Innovation, and Infrastructure **3.**SDG 11: Sustainable Cities and Communities **4.**SDG 13: Climate Action

Technologies used: **1.** Arduino microcontroller **2.** Temperature Sensor (e.g., LM35, DHT11**) 3.**PWM Fan Controller **4.**LCD Display **5**. Power Supply **6**. Resistors and Capacitors **7**. Wiring and Connectors

Group Details:

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Final Approval

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